

ENFIRCE: Environment- and Friction-informed Rod Contact Estimation

Abstract—Sparse shape observations may admit several external-load explanations when a continuum robot experiences body contacts and an independent tip load. We present ENFIRCE, a joint-window maximum-a-posteriori estimator combining measured bending curvature, uncertain environment geometry, and friction history. Every time retains nonlinear three-dimensional Cosserat equilibrium. Contacts on the same plane share its latent parameters at each time, and friction displacement compares the same material coordinate in successive equilibria. We evaluate two fixed two-state multicontact configurations using three sensor seeds, three noise levels, and six factor settings: 108 full-window runs, followed by 36 matched point/Gaussian literature adaptations on byte-identical observation packets. At nominal noise, seed-mean contact-vector RMSEs are 0.215 and 0.288 N; corresponding point-load errors are 3.142 and 2.491 N. Contact, tip, and resultant metrics, review states, branch-conditional uncertainty, and computational costs are reported separately. Shared derivatives reduce duplicated mechanics evaluations while preserving the estimate within a recorded numerical tolerance. The controlled simulations quantify individual force separation under repeated sensor noise, with inspectable factor ablations and branch-conditional diagnostics.

I. INTRODUCTION

Continuum robots can navigate confined spaces, but contact may occur where force sensors cannot readily be placed. A calibrated mechanical model can infer loading from sparse shape measurements. Recovering the resultant force, however, does not necessarily recover its sources. A distal task load and several body reactions may produce similar measured deformations, leaving individual forces poorly separated even when shape residuals are small.

FBG observations have supported continuum-manipulator force sensing [1]. Point-force curvature fitting [2], distributed-load parameterization [3], and probabilistic shape/load inference [4] provide established approaches. Mixed geometric constraints have also been used for multicontact estimation [5]. Combining shape and geometry is therefore not itself a new contribution. Our narrower objective is to combine uncertain environment geometry and friction history for separating body reactions from an independent tip force, while retaining the equilibria of previous and current configurations.

The formulation integrates four components: nonlinear three-dimensional balance at every time; shared latent geometry whose likelihood is counted once per surface and time; friction based on the current material coordinate in the preceding equilibrium; and explicit diagnostics separating numerical feasibility from observation fit and force identifiability. These components preserve the project objective of estimating force magnitude and decomposition from shape and environment information.

The evaluation retains independent truth equilibria, observation packets, raw estimates, and source records. A

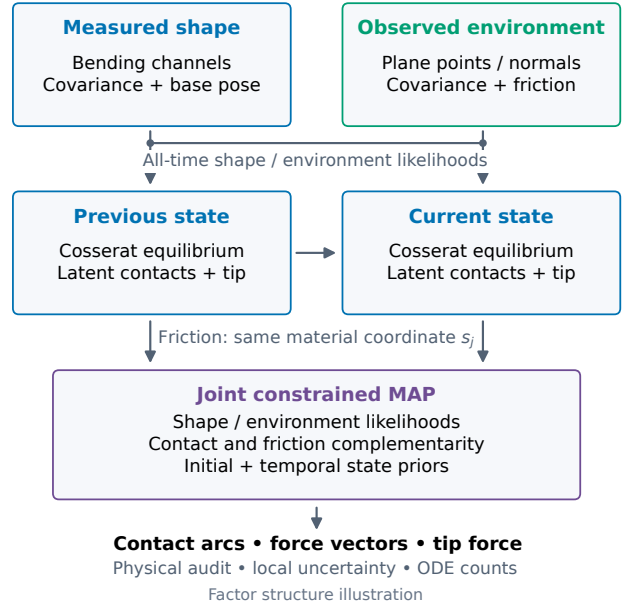


Fig. 1. Factor structure illustration. Previous and current shapes are latent equilibria. Measured bending channels and environment likelihoods enter a joint MAP; friction compares the same material coordinate across times. Forces and diagnostics are estimator outputs.

completed protocol runs six full-window factor settings over two configurations, three sensor seeds, and three noise levels, then evaluates matched curvature-only adaptations. Separate historical density and mismatch experiments expose failure boundaries. Figure 1 summarizes the estimator; Fig. 2 shows actual solved configurations. The implementation is an offline simulation method.

II. RELATED WORK AND COMPARISON SCOPE

Xiao and Chen fit curvature to point-load mechanics [2]; Aloï *et al.* fit shape using transverse Gaussian loads [3]. Ferguson *et al.* formulate a probabilistic batch along arc length [4], distinct from our window over physical times. Tension-profile sensing [6] and mixed geometric/actuation constraints [5] provide different observation sets. Their hardware errors cannot be ranked against our synthetic RMSE without matching inputs, force scales, and metrics.

Recent Gaussian-parameterized graphs fuse strain, tendon tension, and pose for multiple contacts [7]. Discrete Cosserat graphs with actuation uncertainty enable efficient estimation across architectures [8]. Neither complete graph is reproduced here. Our Gaussian curvature baseline adapts Aloï’s parameterization, not these graphs or Ferguson’s Gaussian-process batch.

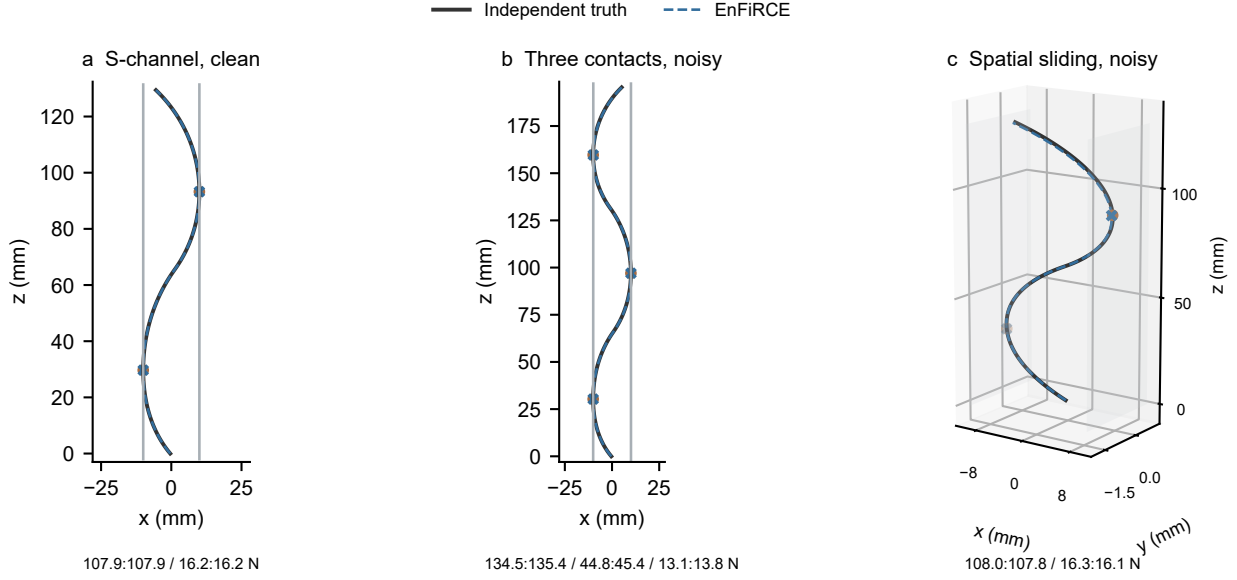


Fig. 2. First solved state from three archived windows: independent truth and EnFiRCE centerlines, true contact dots and estimated crosses. Grey lines/patches visualize clipped portions of the infinite plane model, not finite obstacles. Each bottom entry is true:estimated contact-force magnitude in N, ordered along the rod. The planar projections and spatial view are actual solved configurations, not assumed contact shapes.

Our two inspectable adaptations retain unknown arcs and an independent tip load, use their own shape initializations, and share only the candidate count inferred from observations by EnFiRCE. This conditional comparison tests load estimation rather than count detection. They have no environment or temporal process likelihood; the difference from EnFiRCE is a complete estimator difference, not the isolated effect of one constraint.

III. JOINT ENVIRONMENT AND CONTACT FORMULATION

A. State and observations

Let $k = 1, \dots, T$ index times and $s \in [0, L]$ denote material arc length. Inputs comprise sparse measured curvature channels, calibrated intrinsic curvature $u_0(s)$ and stiffness $B(s)$, base poses, and P observed half-space planes with covariances and friction coefficients. Internal units are mm, N, and rad; B has units N mm^2 . Two measured bending channels remain two: torsion is mechanically predicted, not supplied as a fictitious third channel.

A shape-driven search generates K ordered candidate slots with surface assignments a_j . The physical state is

$$x_k = [\{b_{p,k}, \eta_{p,k}\}_{p=1}^P, \{s_{j,k}, f_{n,j,k}, \beta_{j,k}, \lambda_{j,k}\}_{j=1}^K, f_{e,k}], \quad (1)$$

where $b \in \mathbb{R}^3$ anchors a plane, $\eta \in \mathbb{R}^2$ parameterizes its unit normal, s_j is contact arc length, f_n is normal force, $\beta \in \mathbb{R}^q$ are nonnegative friction-ray coefficients, λ is a slip slack, and $f_e \in \mathbb{R}^3$ is tip force. Contact position is derived as $p_k(s_{j,k})$. A lifted base moment $m_{0,k} \in \mathbb{R}^3$ initializes integration and is absent from physical-state priors. Numerical dimension is $5P + (q + 3)K + 6$ per time; default $q = 4$ gives 30 coordinates for two planes and two slots.

Normals use a two-dimensional spherical chart about the first observed normal and transported tangent generators $D_{p,k}$. Contact force is

$$f_{j,k} = n_{a_j,k} f_{n,j,k} + D_{a_j,k} \beta_{j,k}, \quad \beta_{j,k} \succeq 0. \quad (2)$$

The configurable polygonal cone approximates Coulomb friction. Slots can be inactive with zero force; a bounded candidate set is not a global topology search.

B. Nonlinear equilibrium at every time

With no nominal distributed body loading,

$$N_k(s) = f_{e,k} + \sum_{j: s_{j,k} > s} f_{j,k}. \quad (3)$$

Each time satisfies the full Cosserat equations

$$p'_k = R_k e_3, \quad R'_k = R_k \widehat{u_k}, \\ m'_k = -R_k e_3 \times N_k, \quad u_k = u_0 + B^{-1} R_k^\top m_k, \quad (4)$$

with known base pose and $m_k(L) = 0$. Moment is in world coordinates, curvature in the material frame. Integration segments at loads and calibration discontinuities. Terminal moment is a hard equality on the lifted state. No prior shape, moment arm, or small-deflection replacement is frozen into the current map.

Define $g_{p,k}(r) = n_{p,k}^\top (r - b_{p,k})$. Nonpenetration is sampled over the whole centerline, including candidate positions. Smooth interior active contacts satisfy

$$f_{n,j,k} n_{a_j,k}^\top R_k(s_{j,k}) e_3 = 0. \quad (5)$$

Force-weighted tangency does not represent end contact or a corner with multiple normals. Geometry is currently infinite half-spaces and centerlines, without finite-patch or rod-radius corrections.

C. Previous equilibrium and friction

Tangential displacement uses the current material coordinate:

$$v_{j,k} = (I - nn^\top)[p_k(s_{j,k}) - p_{k^-}(s_{j,k})], \quad (6)$$

where k^- is the packet's actual predecessor and $n = n_{a_j,k}$. Its shape is evaluated at the *same* $s_{j,k}$ rather than at the preceding contact coordinate. This matters as contact migrates along a bent rod.

Let $\gamma = \mu_{a_j,k} f_{n,j,k} - \mathbf{1}^\top \beta_{j,k}$ and $w = D_{a_j,k}^\top v_{j,k} + \lambda_{j,k} \mathbf{1}$. Complete friction conditions are

$$\begin{aligned} 0 &\leq g_{a_j,k}(p_k(s_{j,k})) \perp f_{n,j,k} \geq 0, \\ 0 &\preceq w_{j,k} \perp \beta_{j,k} \succeq 0, \\ 0 &\leq \lambda_{j,k} \perp \gamma_{j,k} \geq 0. \end{aligned} \quad (7)$$

They retain normal contact, raywise friction response, and cone saturation. If no predecessor exists, only the static cone is checked and active frictional contact is marked for review. Missing history is not interpreted as measured zero slip.

D. Joint MAP and shared geometry

Let Π select measured curvature axes and $W_{u,k}$ whiten their covariance. A plane observation $z_{p,k} = [\tilde{b}_{p,k}; \tilde{n}_{p,k}]$ has a six-component covariance whitened by $W_{p,k}$. The objective is

$$\begin{aligned} \hat{X} = \arg \min_X \frac{1}{2} \sum_k \|W_{u,k}(\Pi u_k - z_{u,k})\|^2 \\ + \frac{1}{2} \sum_{k,p} \|W_{p,k}([b_{p,k}; n_{p,k}] - z_{p,k})\|^2 \\ + \frac{1}{2} \|W_0(x_1 - \bar{x}_0)\|^2 \\ + \frac{1}{2} \sum_{k=2}^T \left\| W_Q \frac{x_k - x_{k-1}}{\sqrt{\delta t_k}} \right\|^2, \end{aligned} \quad (8)$$

subject to equilibrium, complementarity, bounds, and nonpenetration. Here $\delta t_k = (t_k - t_{k-1})/t_{\text{ref}}$. Each surface likelihood occurs once per time regardless of contact multiplicity. Contacts at one time share geometry; the process permits changes between times. Lifted moments are excluded from priors.

The initial prior is independent of the likelihood. Shape reconstruction, moment fitting, and previous numerical states are initializers rather than extra posterior factors. Full positive-definite initial/process covariances may be supplied. Weak initial geometry priors avoid counting the first camera observation twice. Legacy predecessor measurements become latent equilibrium times, not unbalanced fixed shapes.

IV. NUMERICAL IMPLEMENTATION

A. Candidates and constrained optimization

Candidates are local plane-distance minima on reconstructed measured-channel shapes. Minima are merged within each surface and time, then associated by material-arc distance to the preceding observation. This avoids duplicating a migrating contact solely because it moves beyond the 10 mm merge distance. The temporal gate scales with rod length and

elapsed time; the default candidate gap is 5 mm and at most four tracks are retained. Truncation, ambiguous association, and corner assignments trigger review. Contact arcs may move over the full rod, subject to endpoint bounds and linear ordering $s_{j+1,k} - s_{j,k} \geq 0.1$ mm; fixed initial partitions remain a disclosed ablation.

Initialization uses sparse-curvature moment fits and trust-region curvature-floor stages $10^{-4}, 10^{-5}, 0$ mm $^{-1}$, with an ordering penalty and projection between stages. Final MAP uses declared covariance and hard ordering. Scaled SQP and an interior-point retry solve product relaxations $\tau \in \{10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}\}$. On an inferred active branch, nonlinear least squares first restores feasibility, with a 10^{-4} -weighted original MAP residual limiting observation-shape drift. The fixed restoration merit is re-evaluated: a worse or nonfinite returned point is rejected in favor of its starting point. The original MAP is then optimized with the selected zero-side equalities; its likelihood and constraints are unchanged. Restoration alone is not MAP convergence. A positive optimization exit and all original, unrelaxed constraints are required for acceptance. Earlier acceptable states are retained if later steps fail; branch choices, restoration decisions, and failed exits remain recorded.

B. Diagnostics and local uncertainty

Outputs separately report equilibrium, centerline gap, complementarity, tangency, solver exit, candidate warnings, and history. Whitenened curvature residuals are compared with a chi-square noise scale under a Bonferroni window correction. Because the fitted data are reused, this is a conservative fit/model diagnostic, not an independent force-accuracy test.

Optional local Laplace covariance uses residual Jacobian J and null-space basis Z of the inferred active constraints:

$$C_y = Z[(JZ)^\top JZ]^\dagger Z^\top. \quad (9)$$

SVD computation avoids explicitly squaring Jacobian conditioning. Unresolved force directions receive infinite standard deviation; measurement-only rank is distinguished from prior-supported rank. The covariance conditions on the candidate set and branch. Mode mixtures and calibration uncertainty are not marginalized. Empirical component coverage is evaluated only for repeated sensor noise on fixed configurations and a local branch.

C. Exact mechanics reuse

Full-vector caching formerly reintegrated all times when differences changed only one frame or a nonmechanical plane anchor. An eight-entry per-time LRU cache now uses

$$[m_{0,k}; \{s_{j,k}\}; \{f_{j,k}\}; f_{e,k}]. \quad (10)$$

Calibration and base pose are fixed per time within one invocation. Only exact equality reuses shape/curvature, without rounding or nearby-state approximation. Geometry and predecessor displacement are recomputed. A normal change affecting world force invalidates the key. ODE calls/hits are recorded and caching can be disabled. A single central-difference sweep of the stacked residuals and

physical constraints supplies objective and constraint gradients, measurement-only information, and active-constraint covariance. Its product rows use $\text{diag}(b)J_a + \text{diag}(a)J_b$. Direct force-vector differentiation avoids another mechanics sweep.

V. SIMULATION AND COMPARISON PROTOCOLS

A. Independent truth and repeated observations

An independent continuous shooting solver generates equilibria before sparse sampling. The inverse receives no true arcs, forces, dense shape, or mode labels. A 210 mm alternating-curvature rod forms three contacts between planes; a spatial two-contact configuration includes out-of-plane loading, $\mu = 0.03$, and tip force $[0.1, 0.3, -0.08]^T$ N. Base sliding by 0.5 mm along $+y$ induces opposing friction. This is continuous sliding, not a stick-to-slip experiment. Archived solved geometry in Fig. 2 illustrates the fixture family.

Bending stiffness is 200700 N mm^2 and torsional stiffness $200700/1.3 \text{ N mm}^2$. Radii 0.455/0.66 mm are calibration metadata, without radius collision correction. Reactions near 134 N arise from this synthetic stiffness and imposed deformation (Fig. 5), not a measured hardware range.

Each configuration has two states with two bending channels at 24 positions. Curvature noise SDs are 0, 2.5×10^{-5} , and $5 \times 10^{-5} \text{ mm}^{-1}$; clean inputs retain a 10^{-7} mm^{-1} likelihood floor. Seeds 11, 23, and 37 provide independent draws within each fixed condition. The same standardized draw is scaled across noise levels and reused across equal-size configurations, enabling common-random-number pairing rather than independent cross-condition repetitions. Clean observations are identical. The 18 packets comprise 12 noisy inputs and six clean controls, not independent trajectories. Plane observations are fixed across draws, with point/normal-component SDs of 0.05 mm/0.001 in the likelihood.

B. Same-curvature literature adaptations

Both baselines use full body-frame equations

$$\begin{aligned} h' &= -u \times h - e_3 \times n, \\ n' &= -u \times n - q_\ell, \quad u = u_0 + B^{-1}h, \end{aligned} \quad (11)$$

where $h = R^T m$, $n = R^T N$, and $h(L) = 0$. Backward integration avoids an inner base-moment shooting loop. Forward integration supplies world forces and shape. Point LS fits three unknown local components per contact, inspired by [2] and generalized beyond its straight-rod specialization. Gaussian LS adapts Aloï's transverse parameterization [3]:

$$q_\ell(s) = \sum_j [a_{x,j}, a_{y,j}, 0]^T \frac{e^{-(s-c_j)^2/(2\sigma_j^2)}}{A_j \sqrt{2\pi}\sigma_j}, \quad (12)$$

with normalization A_j over $[0, L]$, estimated width in $[0.25 \text{ mm}, L/3]$, and world force $\int R q_{\ell,j} ds$. Its position likelihood is replaced by the same measured curvature; finite width and transverse loading differ from point-contact truth.

Each baseline restricts ordered arcs to uniform partitions and uses its own 25%, 50%, and 75% initializers within them, 80 iterations per start, and the preceding baseline solution

TABLE I
MATCHED CURVATURE PACKETS AT NOISE SD $2.5 \times 10^{-5} \text{ mm}^{-1}$.
ENTRIES ARE SEED MEANS (N); THREE SENSOR DRAWS PER FIXED
TWO-STATE TRAJECTORY.

Scene	Method	Contact	Tip	Total
Three	EnFiRCE	0.215	0.299	0.341
Three	Point LS	3.142	2.469	2.147
Three	Gaussian LS [†]	0.780	0.989	1.004
Spatial	EnFiRCE	0.288	0.251	0.311
Spatial	Point LS	2.491	0.590	2.655
Spatial	Gaussian LS [†]	0.607	0.681	0.710

[†]Includes a nonpositive final exit; estimates are retained.

as another initializer. No EnFiRCE posterior is supplied. Candidate count alone is shared from observations, so comparison is conditional on count; these fixtures do not cross the baseline arc partitions. Baselines lack environment/process likelihoods; the experiment compares complete estimators, not one isolated factor. Weak amplitude/ 10^6 regularization remains. Local body/tip bounds are $\pm 1000/\pm 10$ N versus EnFiRCE normal/world-tip bounds of $200/\pm 25$ N. These do not clip truth. All 36 baseline runs, including review and nonpositive exits, are retained.

C. Factors, scoring, and provenance

Six settings independently cold-start the same packets: full method; no temporal process prior; static friction cone without displacement complementarity; no gap/tangency/nonpenetration factors; no camera likelihood; and legacy fixed arc partitions. Other factors and declared priors remain unchanged. Removing geometry retains contact normal/friction parameterization; removing camera retains geometric constraints and observation-derived initialization. There are 108 full-window calls, including local covariance.

Contact-vector RMSE is $\sqrt{(KT)^{-1} \sum_{j,k} \|\hat{f}_{j,k} - f_{j,k}\|^2}$, using ordered active-contact matching. Tip and total metrics are vector RMSE; total is $f_e + \sum_j f_j$. Magnitude MAE and arc RMSE are separate. A count mismatch receives no fabricated correspondence. Every input/truth pair has identical bytes across matched methods; truth is loaded only after inference. Source/dependency hashes, MAT/CSV estimates, completion state, and figure source data accompany the results.

VI. RESULTS

A. Multicontact force separation

All 108 factor and 36 adapted-baseline calls completed without inference exceptions. Final nonpositive exits occurred in 3 factor and 2 baseline windows; corresponding review totals were 48 and 8 states, retained rather than filtered. Clean full-method contact RMSE ranged from $7.49\text{e-}07$ to $9.11\text{e-}07$ N: numerical consistency with model-consistent truth, not sensor accuracy.

At nominal noise, mean contact RMSE was 0.215 N for three contacts and 0.288 N for spatial sliding. Point LS gave 3.142/2.491 N; Gaussian LS gave 0.780/0.607 N. EnFiRCE's

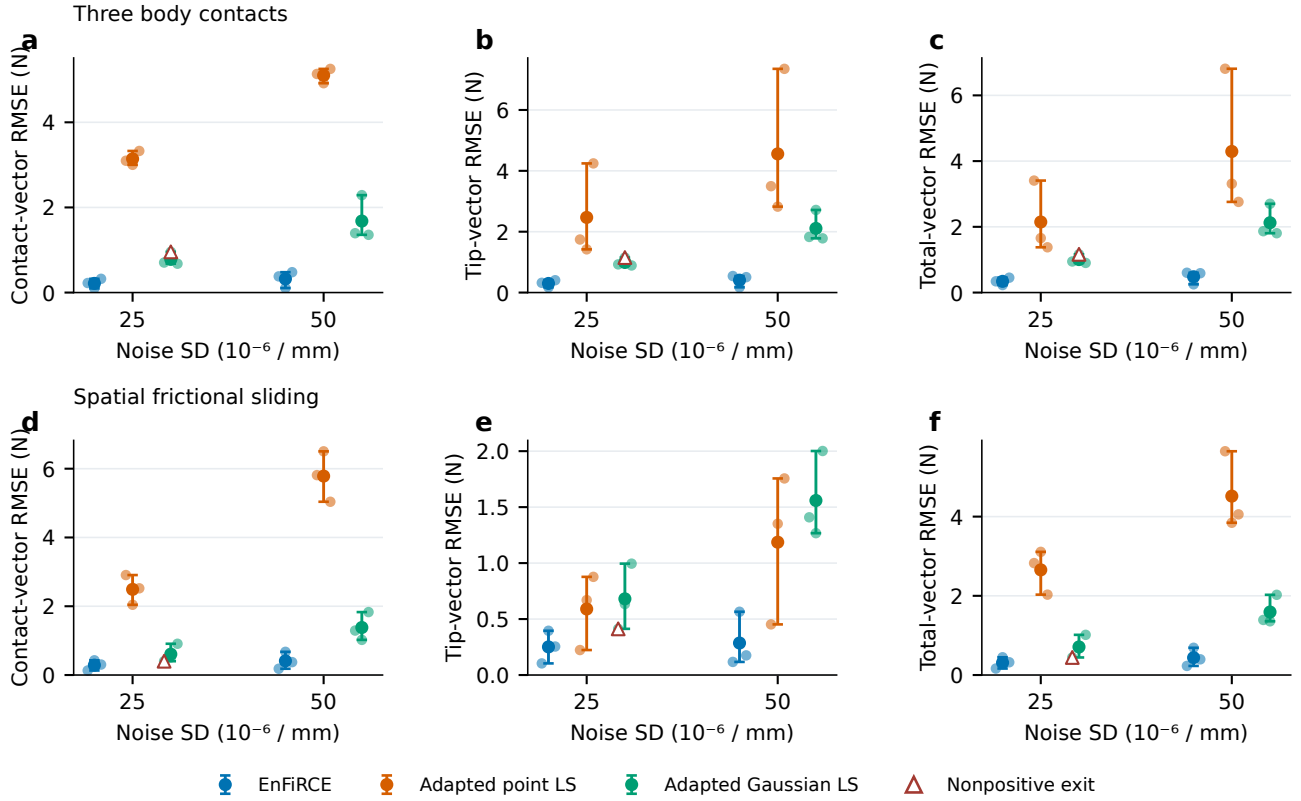


Fig. 3. Matched observation bytes: three independent curvature draws per noise level on two fixed two-state configurations. Dots show individual window RMSEs; markers and whiskers show means and seed min/max, not confidence intervals. Hollow triangles mark nonpositive final exits, retained in the means. Point/Gaussian methods are disclosed curvature-only adaptations, conditioned on observation-derived count.

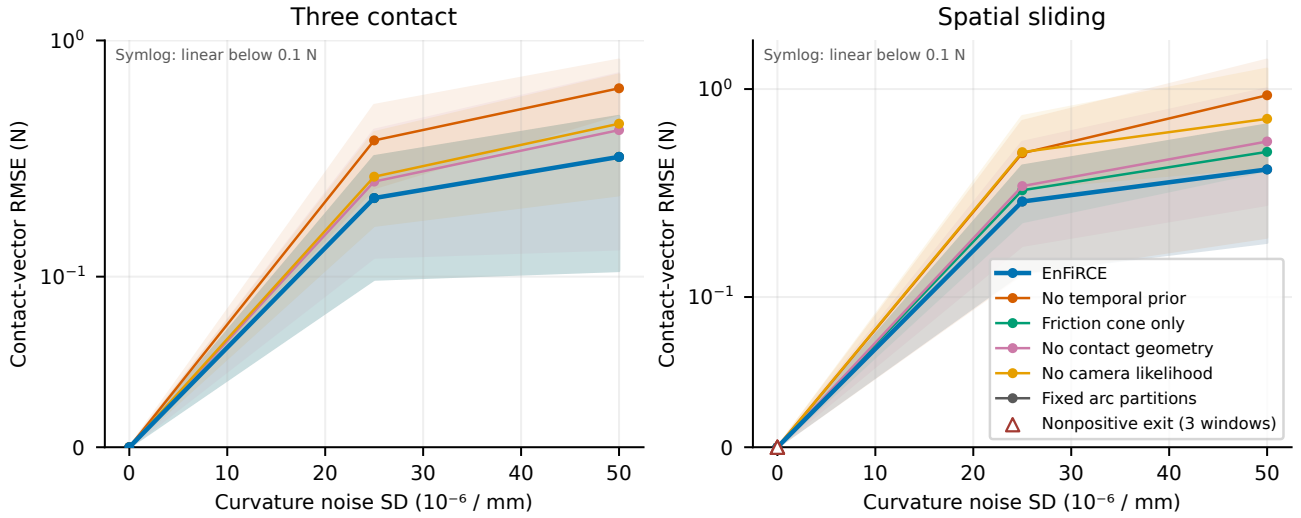


Fig. 4. Full nonlinear window ablations: three sensor seeds per condition, including identical clean controls. Lines show mean contact RMSE and shading shows seed min/max. The symlog ordinate is linear below 0.1 N; crosses mark observation-fit warnings and hollow triangles mark nonpositive final exits (coincident controls counted in legend). All variants independently initialize from observations. Geometry/camera ablations retain their stated remaining priors and parameterizations.

mean was lower in 4/4 contact comparisons and 4/4 resultant comparisons. Figure 3 shows all noisy seed errors rather than only favorable averages. Component errors can cancel in the resultant, so total-force accuracy does not certify individual

contact separation. Force magnitudes and their actual scale appear in Fig. 5.

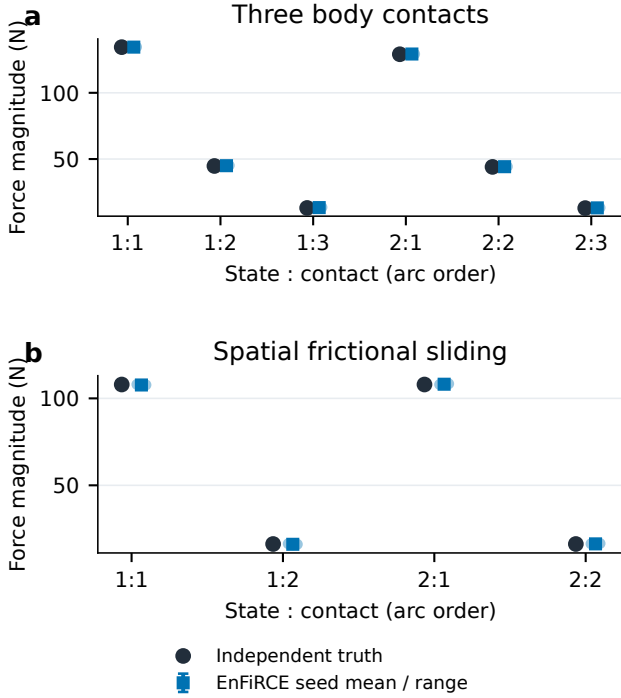


Fig. 5. Actual force magnitudes at nominal noise: independent truth and EnFiRCE seed mean/range for each state and ordered contact. Large synthetic reactions follow the stated calibration.

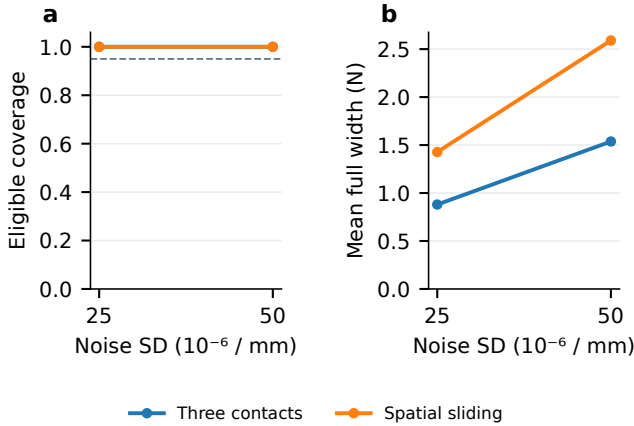


Fig. 6. Branch-conditional intervals: eligible force-component coverage and mean full width. Unresolved components and deterministic clean controls are excluded. Correlated components do not constitute independent coverage trials.

B. Factor ablations and local uncertainty

Figure 4 isolates the specified factor changes on identical packets. Nominal three-contact/spatial mean errors were no temporal prior: 0.378/0.492 N; cone only: 0.215/0.326 N; no contact geometry: 0.253/0.341 N; no camera likelihood: 0.265/0.497 N. These controls do not establish uniform benefit: a removed factor may tie or outperform the full method on an individual configuration. Legacy partitions are inactive when contacts stay inside them, so a tie cannot validate migration performance. A separate engineering

regression checks the corrected search domain under cross-partition migration, without claiming an additional physical experiment.

For noisy full-method windows, 252/252 eligible world-force components fell inside local 95% intervals with mean full width 1.551 N; 0 components were unresolved. Infinite standard deviations do not count as coverage successes. Clean controls are excluded. Components and adjacent times are correlated, and geometry observations are fixed, so this is branch-conditional evidence under sensor noise, not calibration over geometry, materials, or contact modes. Full-method windows retained 9 review states, including 9 spatial initial states lacking predecessor friction observations. They use the static cone and retain the missing-history warning.

C. Failure boundaries and historical controls

Separate ordered planar experiments retain their historical implementation: 396 estimates vary density, noise, and geometric constraints; 54 replay uncertain plane offsets. At nominal noise and 24 positions, pooled contact RMSE was 0.490 versus 2.967 N for its shape-only baseline; eight positions gave 13.576 N. A 1 mm plane error gave 31.092 N under fixed geometry versus 1.341 N with a latent offset SD of 1 mm. These results support sensing/calibration boundaries, not statistics for the full 3-D method.

Corrected independent mismatch truth gave total errors of 17.834/15.964 N when a wall was hidden, and 1.647 N for true/inverse $\mu = 0.03/0.01$. Noise-scaled fit checks flagged both states in each test. A warning neither recovers an absent reaction nor guarantees mismatch detection. Historical three-contact noise also gave a resultant counterexample: EnFiRCE 0.820 N versus Point LS 0.632 N. That distinct input remains available. Tip-only tests infer zero body candidates and a 0.156205 N tip force with model-consistent RMSE 6.78×10^{-10} N.

D. Measured computational cost

The host was 13th Gen Intel Core i9-13900HK (14 physical cores, 20 logical processors), 31.69 GiB visible RAM, Windows 11; MATLAB 24.1.0.2537033 (R2024a). Full-method calls including covariance had median 138.47 s and range 114.65–311.93 s. Factors used six independent scene/seed processes, each with one computational thread; 0 earlier completed calls retain sequential timings. These shared-resource wall times are not isolated latency rankings. The 36 baseline calls retain their sequential single-thread timings. After every worker exited, a sequential clean two-contact derivative experiment used MATLAB default threads and kept mechanics, cold starts, covariance, and central-difference steps identical. Archived regression replay used its original default-thread mode and unchanged tolerances. Separate/shared derivatives took 37.63/31.79 s and 6817/6060 ODE evaluations; maximum force/arc changes were 8.28e-15 N/7.11e-15 mm. One fixed-order trial does not establish timing significance or real-time operation. An earlier exact-cache ablation, without covariance, reduced 116.29 to 50.75 s with zero force difference.

VII. DISCUSSION AND CONCLUSION

EnFiRCE separates body reactions and an independent tip load through full equilibria, shared uncertain planes, and previous-equilibrium friction. Repeated noisy observations and factor controls support bounded multicontact comparisons with the two disclosed adaptations. They do not reproduce official probabilistic graphs or establish state-of-the-art performance. Additional observations in published systems preclude ranking their reported errors against these simulations.

Finite candidates and temporal association may miss reactions or confuse identities. Full-length ordered arcs remove an initialization restriction without enumerating topology. Infinite half-spaces omit curved/finite surfaces and rod-radius corrections. Gravity, shear, extension, material/actuation uncertainty, mode mixtures, and an independent stick-to-slip event lie outside the present nominal evidence. Local covariance and review flags remain diagnostics.

The present experiments assess two fixed full-window configurations under repeated sensor noise. Broader validation should include independent trajectories, informative friction transitions, and input-matched official implementations. Recursive marginalization and analytical sparse sensitivities may improve computational cost. Hardware validation requires calibrated FBG channels, synchronized geometry, measured stiffness, and independent contact/tip force truth.

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